

RECOGNITION STAGE — the contribution of this study
trained on 837 SwSL clips; 604,551 parameters; no external pre-training

SYNTHESIS STAGE — pretrained, used as released
facebook/mms-tts-swh; no fine-tuning; not evaluated here

Video
input



MediaPipe
Holistic



60 x 258
sequence



BiLSTM
128 + 64
(bidirectional)



Temporal
attention
Eqs. (11)-(13)

gesture label



Label to
Swahili text
(lookup)



MMS-VITS
waveform
+ playback

confidence gate
(Algorithm 1)



softmax over 7
classes + threshold

Accuracy reported in this paper (97.62% over 1 seeds) belongs to this stage only.